

MINIMIZING THE NUMBER OF STATES IN A FINITE AUTOMATON

Two strings x and y are L -distinguishable if for some string z , exactly one of the two strings xz, yz is in L .

Two states p and q are distinguishable (or not equivalent i.e. $p \not\equiv q$) if and only if, for some string z , **exactly one** of the states $\delta^*(p, z), \delta^*(q, z)$ is in A .

For every pair (p, q) with $p \neq q$, if exactly one of the two states is in A , then $p \not\equiv q$

For every pair (r, s) of distinct states, if **there is a symbol** $\sigma \in \Sigma$ such that the pair $(\delta(r, \sigma), \delta(s, \sigma))$ comprises of distinguishable states, then r and s are also distinguishable (i.e. $r \not\equiv s$).

The Algorithm:

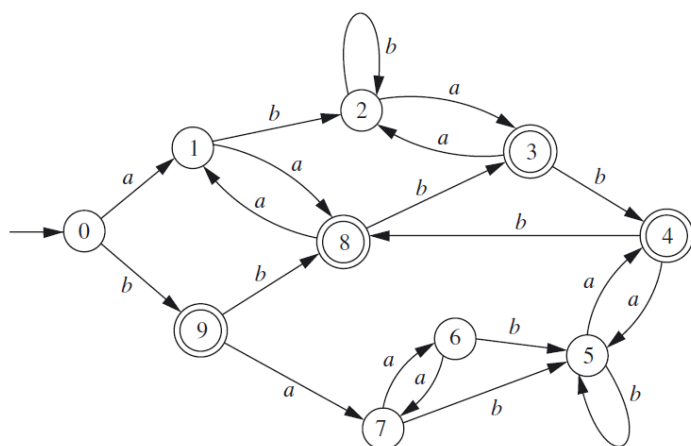
List all unordered pairs of distinct states (p, q) . Make a sequence of passes through these pairs as follows.

On the **first pass**, mark each pair (p, q) for which exactly one of the two states p and q is in A .

On each subsequent pass, examine each **unmarked** pair (r, s) ; if there is a symbol $\sigma \in \Sigma$ such that $\delta(r, \sigma) = p, \delta(s, \sigma) = q$, and the pair (p, q) has already been marked, then mark (r, s) .

After a pass in which no new pairs are marked, stop. At that point, the marked pairs (p, q) are precisely those for which $p \not\equiv q$.

When the Algorithm terminates, every pair (p, q) that **remains unmarked** represents two states that **can be combined** into one. It may happen that every pair ends up marked; this means that the FA already has the fewest states possible.



(a)

1	2								
2	2								
3	1	1	1						
4	1	1	1						
5	2			1	1				
6	2	2	2	1	1	2			
7	2	2	2	1	1	2			
8	1	1	1			1	1	1	
9	1	1	1	2	3	1	1	1	2
	0	1	2	3	4	5	6	7	8

(b)

